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**EX-504 (GS)**  
**B.E. V Semester** Examination, June 2020  
**Grading System (GS)**  
**Power Electronics Devices and Circuits**  
**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. A single phase voltage controller with an R-L load is connected to a 110 volts source if  $R=10$  Ohms and  $L = 20\text{mH}$  and  $\alpha = 90^\circ$ . Find
  - i) The Rms output current
  - ii) The power delivered to the load
  - iii) The power factor
2. A single phase half wave circuit is fed by 240 V rms ac supply. Assume that load current is continuous and voltage drop across the thyristor when ON is 1.5 V. Find average load voltage for firing angle of  $0^\circ$  and  $30^\circ$ .
3. Explain the applications of the following devices.
  - i) GTO
  - ii) SCR
  - iii) MOSFET
  - iv) IGBT
4. What are the drawbacks of resistive triggering circuit? How it can overcome using R-C triggering circuit? Explain the working of R-C triggering circuit with neat diagram.
5. What is the effect of freewheeling diode on the performance of rectifier circuits? Explain in detail.
6.
  - a) Explain the principle of chopper operation. Write down the various control strategies in chopper.
  - b) How Morgan Chopper works? Write its disadvantages.

OR

What do you mean by PWM? Explain the working principle of any two types of PWM techniques.
7. Explain the configuration and operating principle of cyclo converter.
8. Write short notes on any two of the following:
  - a) AC voltage controller
  - b) Harmonic reduction techniques
  - c) Commutation techniques
  - d) Power diodes and power transistor

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